

Integration — Lesson 7 Test: The Fundamental Theorem of Calculus

10 questions · show all three FTC steps (find F, plug in, subtract) · calculators allowed

Name: _____ Date: _____ Score: _____ / 10

1. Use the Fundamental Theorem to evaluate $\int_0^4 5 \, dx$, then check your answer with a rectangle.

1 pt

2. Use the Fundamental Theorem to evaluate $\int_0^3 2x \, dx$, then check your answer with a triangle.

1 pt

3. Evaluate $\int_0^3 x^2 \, dx$. (Verify your antiderivative by differentiating it before you plug in!)

1 pt

4. Evaluate $\int_1^2 x^3 dx$. Give the exact fraction and a decimal.

1 pt

5. Evaluate $\int_0^2 (3x^2 + 4) dx$.

1 pt

6. Evaluate $\int_1^5 (2x + 1) dx$ with the FTC, then verify your answer with the trapezoid method (heights at the two ends, average, times width). 1 pt

7. A student evaluates $\int_1^3 2x dx$ like this: "Antiderivative of $2x$ is x^2 . Plug in 3: $3^2 = 9$. Answer: 9." Find the mistake, explain it in one sentence, and give the correct answer.

1 pt

8. When we use the FTC, we drop the "+C" from the antiderivative. Show with one line of algebra why $(F(b) + C) - (F(a) + C)$ makes the C harmless, no matter what number C is. 1 pt

9. A pump's flow rate is $r(t) = 8t$ liters per hour (it speeds up over time). Use the FTC to find the total water pumped from $t = 1$ to $t = 2$ hours. Include units. 1 pt

10. Challenge: evaluate $\int_0^3 (x^2 - x + 2) dx$. Work term by term and keep your fractions tidy. 1 pt
